Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 104502

Subject: Digital Signal Processing

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The output of an LTI system can be determined using
(i) Sampling (ii) Convolution
(iii) Quantization (iv) Modulation

(b) Which of the following is an advantage of DSP over analog systems?
(i) More noise (ii) Less flexibility
(iii) Better accuracy and reliability (iv) Difficult storage

(c) A system is said to be LTI if it is
(i) Linear and Time Invariant (ii) Linear and Time Variant
(iii) Logical and Time Invariant (iv) Linear and Tuned

© Apna BEU

(d) Convolution is used to determine
(i) Frequency response (ii) System output
(iii) Sampling frequency (iv) Fourier coefficients

(e) The DTFT of a discrete-time signal is
(i) Always periodic (ii) Non-periodic
(iii) Finite (iv) Real only

(f) Aliasing occurs when the sampling frequency is
(i) Greater than Nyquist rate (ii) Equal to Nyquist rate
(iii) Less than Nyquist rate (iv) Double the signal frequency

(g) FFT is mainly used to
(i) Increase noise (ii) Reduce computational complexity of DFT
(iii) Amplify signals (iv) Sample analog signals

(h) Correlation is used to measure
(i) Power consumption (ii) Similarity between signals
(iii) Frequency distortion (iv) Sampling error

(i) The DFT converts a signal from
(i) Time domain to frequency domain (ii) Frequency domain to time domain
(iii) Analog to digital (iv) Continuous to discrete amplitude

(j) Nyquist sampling theorem states that sampling frequency should be
(i) Equal to signal frequency (ii) Twice the maximum frequency
(iii) Half the maximum frequency (iv) Four times the signal frequency

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2 (a) Explain the basic elements of a Digital Signal Processing system with block diagram. [7]

(b) Discuss the advantages of DSP over analog signal processing systems. [7]

Q.3 (a) Classify signals into continuous-time and discrete-time signals with examples. [7]

(b) Define Linear Time Invariant (LTI) systems and explain their properties. [7]

Q.4 (a) Explain convolution sum and determine the response of an LTI system. Use: [7]

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

(b) Describe discrete-time systems using difference equations. [7]

Q.5 (a) Define Discrete Time Fourier Transform. [7]

(b) Explain the properties of DTFT with examples. Use: [7]

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

Q.6 (a) Explain frequency domain representation of LTI systems. [7]

(b) Derive the mathematical expression of DFT. Use: [7]

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}$$

Q.7 (a) A voice signal contains frequencies up to 4 kHz. Determine the Nyquist sampling rate. [7]

(b) Discuss radix-2 Decimation in Time FFT algorithm. [7]

Q.8 (a) Compare DFT and FFT with respect to computational efficiency. [7]

(b) Find the 4-point DFT of the sequence: [7]

$x[n] = \{1,0,1,0\}$

Q.9 Write short notes on ***any two*** of the following: **[7 x 2 = 14]**

(a) FIR filter design

(b) IIR filter design

(c) Discrete time sinusoidal signals.

(d) Frequency domain representation of LTI Systems

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play